# Essential singleton Seidel switches and a construction from arbitrary graphs

Haobo Ma
ChronoAI Pte Ltd; The Omega Institute
Wenlin Zhang
National University of Singapore; The Omega Institute

**Abstract**

A graph is covered if every vertex belongs to an identity Seidel switch consisting of a singleton or the endpoints of an edge. We prove that, in a covered graph of order $2h+1$, $h \geq 1$, with degrees in $\{h, h+1\}$, every vertex requiring a singleton switch is the unique singleton-switch vertex and is fixed by every isomorphism from its switch to the original graph. From each graph $F$ on at least two vertices without isolated vertices we construct a covered graph on $2|V(F)|+1$ vertices with exactly one essential singleton, and determine all its singleton and edge switches. Nonisomorphic base graphs give nonisomorphic constructed graphs; the construction starts with the house. An exhaustive check of orders three through eight finds seven covered graphs with essential singletons; each has a unique singleton-ISS vertex, fixed by every switching isomorphism. The same behavior holds in the sixteen essential-singleton cases at order nine.

## 1 The fixing question

Let $G = (V, E)$ be a finite simple graph. For $S \subseteq V$, its Seidel switch $G^S$ toggles precisely the adjacencies between $S$ and $V \setminus S$. Following Gervacio [1], call $S$ an *identity Seidel switch* (ISS) if $G^S \cong G$. A singleton ISS is a vertex-ISS; an edge-ISS consists of two adjacent vertices. A graph is *covered* if every vertex lies in a singleton ISS or an edge-ISS. Set

$$W(G) = \{v : \{v\} \text{ is an ISS}\}, \qquad E(P(G)) = \{xy \in E(G) : \{x, y\} \text{ is an ISS}\},$$

and let $U(G)$ be the set of isolated vertices of $P(G)$. In a covered graph, $U(G) \subseteq W(G)$, and these vertices require singleton switches. All switching isomorphisms below have direction $G^S \longrightarrow G$.

Gervacio [2] proposed studying $U(G)$ and asked whether in a covered graph every $v \in U(G)$ admits a switching isomorphism fixing $v$, suggesting $|U(G)| = 1$ as a first case and the house as the initial example. He also gave the exact criterion that such an isomorphism exists if and only if some automorphism of $G - v$ exchanges $N_G(v)$ and $V \setminus (\{v\} \cup N_G(v))$. Indeed, restriction gives the required automorphism, and extension by $v \mapsto v$ gives the converse.

We answer the question affirmatively when the degrees lie in $\{h, h+1\}$, where $|V| = 2h+1$, and construct a family with $|U(G)| = 1$ from every graph on at least two vertices without isolated vertices. The question for arbitrary covered graphs, including arbitrary covered graphs with $|U(G)| = 1$, remains open. Our arguments use switching covariance and exact degree changes; they do not assume that the ISS subsets form a group.

## 2 Tracking singleton switches

For a permutation $p$ of $V$, write $pG$ for the relabelled graph, so that $p(G^S) = (pG)^{p(S)}$. Also $(G^S)^T = G^{S \triangle T}$. These identities concern labelled graphs and hold for all subsets.

**Lemma 1.** *Suppose $v \in W(G)$ and $p : G^{\{v\}} \to G$ is an isomorphism. Then $p(v) \in W(G)$. In particular, if $W(G) = \{v\}$, every such $p$ fixes $v$. Moreover,*

$$W(G^{\{v\}}) = \{v\} \cup \{x \neq v : \{v, x\} \text{ is an ISS of } G\}. \tag{1}$$

*For every such $v$ and $p$, with $k = |W(G)|$,*

$$d_{P(G)}(v) = k - 1 - d_{G[W(G)]}(p(v)). \tag{2}$$

*In particular, if $v \in U(G)$, then $p(v)$ is adjacent to every other vertex of $W(G)$. Consequently, if $W(G)$ is independent and $v \in W(G) \cap U(G)$, then $W(G) = \{v\}$ and every switching isomorphism at $v$ fixes $v$.*

*Proof.* Covariance gives $(pG)^{\{p(v)\}} = G$, hence $pG = G^{\{p(v)\}}$. Thus $G^{\{p(v)\}} \cong G$, proving the first claim. Since $G^{\{v\}} \cong G$, for $x \neq v$ the singleton $x$ is an ISS of $G^{\{v\}}$ exactly when $G^{\{v,x\}} \cong G$, which proves (1); switching $v$ a second time returns $G$.

If $v \in U(G)$, every $x \neq v$ on the right of (1) is a nonneighbor of $v$ in $G$, and therefore a neighbor in $G^{\{v\}}$. The isomorphism $p$ takes $W(G^{\{v\}})$ onto $W(G)$, so $p(v)$ is adjacent to every other vertex of $W(G)$. In general, the edge-ISS partners of $v$ in $G$ are precisely its nonneighbors in $W(G^{\{v\}}) \setminus \{v\}$, which proves (2). An independent graph with a vertex adjacent to all its other vertices has just one vertex. □

The distinction between $|U(G)| = 1$ and $|W(G)| = 1$ matters: the lemma proves the fixing conclusion from the latter hypothesis. It does not identify the two conditions in arbitrary covered graphs. If $G[W(G)]$ is complete, (2) gives $U(G) = W(G)$ for a covered graph. Hence a covered graph with $|U(G)| = 1$ and $|W(G)| > 1$ must have $|W(G)| \geq 3$, and its induced singleton graph can be neither complete nor independent.

Switching a singleton changes the edge count by $|V| - 1 - 2d_G(v)$. Switching an adjacent pair $xy$ changes it by $2(|V| - d_G(x) - d_G(y))$. Thus

$$v \in W(G) \implies 2d_G(v) = |V| - 1, \qquad xy \in E(P(G)) \implies d_G(x) + d_G(y) = |V|. \tag{3}$$

In particular a covered graph with $U(G) \neq \varnothing$ has odd order.

**Theorem 2.** *Let $G$ have order $2h + 1$, where $h \geq 1$, and suppose every degree is $h$ or $h + 1$. If $v \in W(G) \cap U(G)$, then $W(G) = \{v\}$ and every isomorphism $G^{\{v\}} \to G$ fixes $v$. Thus Gervacio's fixing question has an affirmative answer for every covered graph with these degrees. Furthermore, in any such graph an edge $xy$ is an edge-ISS if and only if every other vertex is adjacent to exactly one of $x, y$.*

*Proof.* Take $v \in W(G)$. By (3), $d_G(v) = h$. A neighbor of $v$ of degree $h$ would have degree $h-1$ after switching, and a nonneighbor of degree $h+1$ would have degree $h+2$. Both are impossible in a graph isomorphic to $G$. It follows that

$$N_G(v) = \{x : d_G(x) = h + 1\}. \tag{4}$$

Every member of $W(G)$ has degree $h$, so $W(G)$ is independent. Lemma 1 gives the fixing assertion.

Now let $xy$ be an edge-ISS. For $z \notin \{x, y\}$, put $c_z = |N_G(z) \cap \{x, y\}|$. Its new degree is $d_G(z) + 2 - 2c_z$. If $c_z = 0$ or 2, this falls outside $\{h, h + 1\}$. Thus $c_z = 1$ for every such $z$. Conversely, if every outside vertex sees exactly one endpoint, the transposition $(x\ y)$, fixing all other vertices, is an isomorphism $G^{\{x,y\}} \to G$. □

## 3 A family with one essential singleton

Let $F$ be any simple graph on $\{1,\dots,m\}$, $m \geq 2$. Construct $C(F)$ on

$$\{v\} \cup A \cup B, \qquad A = \{a_1,\dots,a_m\}, \quad B = \{b_1,\dots,b_m\},$$

as follows. The vertex $v$ is adjacent to all of $A$ and none of $B$. Both $A$ and $B$ induce copies of $F$. Between the copies, put

$$a_i b_j \in E(C(F)) \quad \Longleftrightarrow \quad i = j \text{ or } ij \notin E(F). \tag{5}$$

For distinct indices the second alternative has its usual meaning; in particular every $a_i b_i$ is an edge.

**Theorem 3.** *For every $F$ as above, $C(F)$ is covered and has degrees*

$$d(v) = m, \qquad d(a_i) = m+1, \qquad d(b_i) = m.$$

*Its singleton ISS vertices are exactly*

$$W(C(F)) = \{v\} \cup \{b_i : d_F(i) = 0\}. \tag{6}$$

*Its edge-ISS are exactly*

$$\{a_i b_j : N_F(i) = N_F(j)\} \ \cup \ \{v a_i : d_F(i) = 0\}. \tag{7}$$

*Consequently,*

$$U(C(F)) = \begin{cases} \{v\}, & \text{if } F \text{ has no isolated vertices,} \\ \varnothing, & \text{otherwise.} \end{cases}$$

*If $F$ has no isolated vertices, every switching isomorphism at $v$ fixes $v$. Nonisomorphic graphs $F$ without isolated vertices yield nonisomorphic graphs $C(F)$.*

*Proof.* The degree formulas follow from $d_F(i) + (m - d_F(i)) = m$. The permutation

$$\tau(v) = v, \qquad \tau(a_i) = b_i, \qquad \tau(b_i) = a_i$$

is an automorphism of $C(F) - v$ exchanging $A$ and $B$: the internal copies agree, and the condition in (5) is symmetric in $i, j$. Extending it by fixing $v$ certifies the singleton switch at $v$. For each $i$, the vertices $a_i, b_i$ are complementary outside their pair, including at $v$. Their transposition certifies an edge-ISS. These switches cover every vertex.

By (3), no $a_i$ is a singleton ISS. If $b_i$ is one, (4) requires $N(b_i) = A$, which holds precisely when $i$ is isolated in $F$. Conversely, for such an $i$, the vertices $v, b_i$ have identical open neighborhoods $A$, so their transposition is an automorphism of $C(F)$. Relabelling the singleton ISS at $v$ therefore gives one at $b_i$. This proves (6).

By Theorem 2, edge-ISS are exactly the adjacent pairs complementary outside their pair. The degree sum in (3) restricts them to $a_i b_j$ or $v a_i$. For an adjacent pair $a_i b_j$ with $i \neq j$, we have $ij \notin E(F)$. For each $k \notin \{i, j\}$, complementarity at $a_k$ (equivalently, at $b_k$) requires $\mathbf{1}_{ik \in E(F)} = \mathbf{1}_{jk \in E(F)}$. At $v, a_j, b_i$ complementarity then holds automatically. Thus the two adjacency rows agree, and conversely their equality gives complementarity everywhere. For $i = j$ complementarity was already checked. This gives $N_F(i) = N_F(j)$; if $i \neq j$ it also implies $ij \notin E(F)$, so the proposed pair is adjacent. For $v a_i$, complementarity requires $a_i$ to have no neighbor in $A \setminus \{a_i\}$ and all of $B$ as neighbors, which is exactly $d_F(i) = 0$. This proves (7).

Every $a_i, b_i$ lies in the edge-ISS $a_i b_i$. Formula (7) decides whether $v$ has an edge-ISS partner, giving the assertion about $U$. If $F$ has no isolated vertices, (6) and Lemma 1 show that every switching isomorphism at $v$ fixes it. Finally, any isomorphism between two such constructed graphs must send their unique singleton-ISS vertices to each other, hence their neighborhoods to each other. Those neighborhoods induce the respective base graphs $F$. □

Taking $F = K_2$ gives the house: $C(F) - v$ is a four-cycle, and $v$ is joined to the endpoints of one of its edges. Taking $F = K_m$ proves existence with $|U| = 1$ at every odd order at least five. More generally, if $a_m$ denotes the number of isomorphism classes of $m$-vertex graphs without isolated vertices, the theorem gives at least $a_m$ covered isomorphism classes of order $2m+1$ with $|U| = 1$.

The degree-sum test does not explain essentiality in this family. When $F$ has no isolated vertices, each candidate $va_i$ passes that test. Choose $j \in N_F(i)$. The vertex $a_j$ is a common neighbor of $v, a_i$, so switching that pair changes its degree from $m+1$ to $m-1$, which is absent from $C(F)$. This explicitly rejects every possible edge partner of $v$.

## 4 Finite catalogues and the remaining question

We recomputed all singleton and adjacent-pair switches for orders three through eight, using the NetworkX distribution [4] of the Read–Wilson atlas [3] through order seven and McKay's catalogue [5] at order eight. These calculations give:

| Order | Input classes | Covered | $U = \varnothing$ | $\lvert U\rvert = 1$ |
|---|---|---|---|---|
| 3 | 4 | 1 | 1 | 0 |
| 4 | 11 | 3 | 3 | 0 |
| 5 | 34 | 3 | 2 | 1 |
| 6 | 156 | 11 | 11 | 0 |
| 7 | 1044 | 10 | 4 | 6 |
| 8 | 12346 | 35 | 35 | 0 |

No covered graph in this range has $|U| \geq 2$. In all seven with $U \neq \varnothing$, the unique essential vertex is also the unique singleton-ISS vertex. Thus Lemma 1 explains why every switching isomorphism there fixes it. The following table gives graph6 labels, the essential vertex (numbered from zero), and one certificate. The tuple lists the images of vertices in order.

| Graph6 | Vertex | Fixing permutation | All isomorphisms |
|---|---|---|---|
| `Dlc` | 4 | $(1, 0, 3, 2, 4)$ | 2 |
| `FcBzo` | 0 | $(0, 2, 1, 4, 3, 6, 5)$ | 4 |
| `FfxbO` | 0 | $(0, 2, 1, 5, 6, 3, 4)$ | 1 |
| `FllHo` | 0 | $(0, 5, 4, 6, 2, 1, 3)$ | 1 |
| `FBfnO` | 0 | $(0, 5, 6, 4, 3, 1, 2)$ | 4 |
| `F`urg` | 2 | $(3, 5, 2, 1, 6, 4, 0)$ | 6 |
| `FHN]w` | 1 | $(2, 1, 0, 5, 6, 3, 4)$ | 4 |

Each successful switch was verified on every vertex pair. Negative isomorphism decisions use exhaustive degree-compatible bijection search. For the seven essential cases, a second check tested all $n!$ permutations and confirmed the displayed counts and the fixing property. Catalogue completeness uses the cited input catalogues. As a separate check of Theorem 3, all 1098 labelled base graphs of orders two through five were constructed and their complete singleton and edge-ISS lists compared with (6)–(7).

An exhaustive supplementary check of McKay's 274668 graphs of order nine gives 33 covered graphs: 17 with $U = \varnothing$ and 16 with $U = W = \{v\}$. In all sixteen, every switching isomorphism at $v$ fixes $v$. No covered graph of order nine has $|U| \geq 2$. At the small orders excluded from the tables, the one-vertex graph has $U = W = V$ and its switch is the identity; the only covered graph of order two is $K_2$, with $U = \varnothing$.

These finite results leave two structural questions. Does every covered graph with $|U| = 1$ have a unique singleton-ISS vertex, or at least a fixing switching isomorphism? Can a covered graph have $|U| \geq 2$, and if so, which graphs can occur as $G[U]$? The construction answers the existence problem for $|U| = 1$ at every odd order at least five, while the fixing question beyond Theorem 2 remains unresolved.

## References


[1] S. V. Gervacio, *On identity Seidel switches*, arXiv:2601.04530v1, 2026. `https://arxiv.org/abs/2601.04530v1`.

[2] S. V. Gervacio, private communication, 12 October 2026.

[3] R. C. Read and R. J. Wilson, *An Atlas of Graphs*, Oxford University Press, 1998.

[4] NetworkX developers, *Graph atlas*, `graph_atlas_g` documentation; atlas data version 3.4.2. `https://networkx.org/documentation/stable/reference/generated/networkx.generators.atlas.graph_atlas_g.html`.

[5] B. D. McKay, *Graphs*, small graph data catalogue. `https://users.cecs.anu.edu.au/~bdm/data/graphs.html`.